

Jesuino Vieira Filho

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Education

MSc in Computer Science (AI Specialization)

Université de Montréal

Sep. 2025 – Present

Montréal, Canada

- Admitted for Fall 2025 intake

BSc in Mechatronics Engineering

Federal University of Santa Catarina

Aug. 2016 – Sep. 2022

Joinville, Brazil

- Average grade: 8.68/10.00 (2nd in class, top 5%)
- Completed two graduate-level courses in Data Science and Machine Learning (Grade: 9.50/10.00)
- Bachelor's thesis: "Comparison of machine learning methods for short-term urban water demand forecasting in a coastal tourist city" (Advisor: Pablo Andretta Jaskowiak; Grade: 10.00/10.00)

Research Interests

representation learning, geometric deep learning, graph neural networks, generative models, ai4science

Professional Experience

Data Scientist & Software Developer

IconPro GmbH

Aug. 2021 – Apr. 2025

Remote

- Developed a library of custom Spark transformers, estimators, and utility functions
- Improved model reliability and explainability with SHAP, conformal predictions, and OOD detection
- Optimized a distributed data pipeline on Kubernetes, reducing runtime by 40%
- Designed a benchmarking tool to evaluate an RL optimization algorithm against traditional methods

Data Scientist & Software Developer (Intern)

IconPro GmbH

Feb. 2020 – Aug. 2021

Aachen, Germany

- Developed a time series analysis and forecasting (TSAF) library for predictive maintenance
- Implemented a statistical process control (SPC) module to monitor manufacturing processes
- Built an ETL pipeline to stream data from manufacturing machines into a database

Research & Teaching Experience

Research Assistant at LISHA

Federal University of Santa Catarina

Aug. 2018 – Jul. 2019

Joinville, Brazil

- Conducted computer vision research, evaluating deep learning models for human detection

Teaching Assistant for Programming II

Federal University of Santa Catarina

Aug. 2017 – Dec. 2017

Joinville, Brazil

- Course covered data structures and sorting algorithms taught in C

Publications († Peer-reviewed, * First author)

† **Vieira Filho J***, Scortegagna A, de Sousa Dias Vieira AP, Jaskowiak PA. Machine learning for water demand forecasting: Case study in a Brazilian coastal city. *Water Practice & Technology*, 1 May 2024, 19 (5), 1586–1602

Vieira Filho J*, Jaskowiak PA. Comparação de métodos de deep learning pré-treinados da biblioteca OpenCV para detecção de pessoas em ambientes internos. *Revista Eletrônica de Iniciação Científica em Computação* 2020, 18 (4)

Further Education

NeuroAI 14 Jul. 2025 – 25 Jul. 2025
Neuromatch Academy Remote

Spark on Kubernetes 17 Oct. 2023 – 18 Oct. 2023
Data Engineering Academy Remote

School on Complex Networks & ML in Neuroscience 22 May 2023 – 26 May 2023
ICTP South American Institute for Fundamental Research São Paulo, Brazil

Writing & Publishing Scientific Papers 12 Jul. 2021 – 20 Aug. 2021
Federal University of Santa Catarina Joinville, Brazil

Honors & Awards

Bourse d'exemption UdeM Sep. 2025
Université de Montréal

Outstanding Academic Performance 1st Semester, 2020
Federal University of Santa Catarina

Institutional Scientific Initiation Scholarship Program (PIBIC) Aug. 2018 – Jul. 2019
National Council for Scientific and Technological Development (CNPq)

Selected Skills

Programming Languages
Python, SQL, C++, C, LaTeX, Shell scripting, MATLAB

Machine Learning Libraries
PyTorch, scikit-learn, Apache Spark, OpenCV, pandas, NumPy

Infrastructure & Development Tools
Docker, Kubernetes, Git, Linux

Languages
Portuguese (Native), English (C1), German (A2), French (A1–A2, in progress)